

Automated EEG Analysis for Harmful Brain Activity Classification



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Abstract In the evolving landscape of medical diagnostics, the ability to swiftly and accurately detect critical brain activities from electroencephalography (EEG) signals is paramount. This research introduces a novel algorithm. Designed to automate the detection and classification of harmful brain patterns such as seizures, which are crucial for timely medical intervention. Manual EEG analysis is resource-intensive and prone to inaccuracies, underscoring the urgent need for automated solutions. Our proposed method offers a streamlined, precise approach, empowering neurologists and researchers to identify seizure-related brain activities more efficiently, ultimately improving diagnostic accuracy and patient care.

1 Introduction

Electroencephalography (EEG) is a powerful tool for monitoring and understanding brain activity by detecting electrical signals produced by neurons. This non-invasive technique has diverse applications, including diagnosing neurological disorders, studying sleep patterns, and even controlling devices through brain–computer interfaces. However, the manual analysis of EEG data poses significant challenges, including time and resource constraints and potential errors resulting from human subjectivity.

In the context of neurological disorders such as epilepsy, timely and accurate detection of abnormal brain activity, particularly seizures, is paramount for effective diagnosis and treatment. Conventional methods of diagnosis depend heavily on expert interrogation of EEG recordings which is very risky and can lead to false interpretation and can as well introduce irregularity among practitioners hence it

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takes a lot of time. The increase in demand for proper and errorless EEG analysis only increases these challenges which highlights the need for automated solutions which can increase and improve the efficiency and productivity.

Our research endeavors to address these limitations. The key motives of this research involve developing an efficient EEG analysis algorithm and an in-depth comparison of different models. Our primary objective with this study is to enhance the speed, accuracy and objectivity of EEG-based diagnostics, so that practitioners can rely on automated solutions and detect whether the patient is having seizures or not and hence save time which can lead to faster detection of critical brain activities. The uniqueness of this algorithm is that it uses advanced CNN models, mainly EfficientNet, which incorporates feature extraction in an efficient manner and also improves classification accuracy which is more accurate than the traditional methods. The main motive of this approach is to target the analysis of EEG patterns which can detect epileptic seizures and thus can help in refining the diagnostic process.

2 Literature Review

The process of automatically detecting and classifying critical medical activities in electroencephalography (EEG) signals has become an important area of research due to its potential to improve clinical practices and neuroscience studies. This review has explored recent developments in this area, focusing on studies addressing the challenges associated with manual EEG analysis highlighting the field's rapid advancement and growth.

Over the past years there has been a significant advancement in developing automated algorithms for analyzing EEG data for classifying brain activities. These algorithms include machine learning techniques like convolutional neural networks (CNNs), recurrent neural networks (RNNs), and support vector machines (SVMs) which help by extracting important features from EEG signals and data provided and classify them into different categories such as seizure and non-seizure. These algorithms are trained on large datasets of labeled EEG recordings (supervised learning) which then shows promising results and improved accuracy in detecting critical brain activities.

A number of studies showed the effectiveness of automated EEG analysis in the clinical department, which gave promising and significant results. For an instance, in a research by Acharya et al. (2018), a CNN-based algorithm achieved a sensitivity of 92.6% and a specificity of 91.3% in identifying brain seizures, which outperformed human experts with its high accuracy which made path for personalized treatment for patients suffering from epileptic seizures.

Nanthini et al. [1] explored methods for analyzing EEG signals to identify neurological irregularities (mainly epilepsy). Their research involves eight statistical features that were mean, median, mode, standard deviation, maximum, minimum, skewness, and kurtosis which were taken from raw EEG data. The support vector machine (SVM) classifier uses this raw data as an input features to create an automatic

system for abnormalities detection. Result: The study was tested into two separate sets both of which gave promising results for automated seizure detection, hence helping to select the relevant set of features for reliable and accurate identification [1].

Zhao et al. [2] published in *Annals of Translational* discusses the requirement and importance of deep learning algorithms in medical imaging. Processes like MRIs and CT scans use deep learning algorithms for tasks like disease detection, image classification, organ segmentation, etc. The result of this research expresses the role of convolutional neural networks (CNNs) and how it helped in increasing precision, accuracy, and efficiency of medical image processing. The paper also addresses some key challenges in this field, including the complexity of medical image segmentation and the difficulties they faced in it, the need for robust datasets, and the importance of using reliable performance metrics to evaluate deep learning models [2].

Balasamy et al. [3] examined a number of CNN-based deep neural networks, and a noteworthy outcome was obtained on ImageNet Challenger, i.e., the most significant image classification and segmentation task. CNN's primary advantage is in its ability to recognize crucial characteristics without the need of human assistance. The result of this review is that the highest performance is achieved by the ResNet architecture among all the CNN neural networks [3].

Santhi et al. (2020) published in *Advances in experimental medicine and biology* in the year 2020 addressed how deep learning has an important influence on medical image analysis, showing how it can perform better than conventional techniques and how it might transform healthcare by enabling more precise and effective medical image processing and diagnosis. The result is that, if effectively designed, verified, and put into practice, effective data analytics from CAD or AI tools if put hand in hand with the human intelligence of clinicians could increase precision and efficiency and hence patient care [4].

Virra and Aizzah et al. (2024) employed transfer learning and data augmentation techniques to train deep learning models utilizing electroencephalography data as input. This review's findings demonstrated that the deep learning model's classification accuracy of harmful brain activity was increased by incorporating augmentation approaches and custom transfer learning EfficientNetV2. The optimal model achieved a validation loss of 0.9603 and a validation accuracy of 69%, indicating moderate performance of the model [5].

Marija Habijan et al. (2024) discussed an ensemble deep learning method for classifying harmful brain activity by using neural networks to increase accuracy in brain signal analysis by using various deep learning models and combining their outputs. The model studied combined the strengths of ResNet1D and Gated Recurrent Unit (GRU) which lead to improvement in robustness and accuracy in classifying critical brain activity patterns. The result of this research suggests that incorporating an ensemble deep learning method helps in increasing accuracy and it also provides a more reliable framework for classifying abnormal brain activities [6].

From the above studies, it is concluded that automatic detection and classification of critical brain activities using EEG signals can change how neurological disorders are diagnosed hence helping the practitioners to rely on those results this is achieved with the help of advanced machine learning techniques and robust and large EEG datasets.

3 Implementation

The proposed algorithm's implementation approach begins with gathering and pre-processing a variety of EEG data, then feature extraction to identify relevant patterns. Then, created and trained the machine learning models for classification, with emphasis on metrics like precision and accuracy. The next step is improving performance by fine-tuning and optimizing the model. Ensuring accessibility, proper documentation and deployment will enable seamless detection and classification of EEG signals. This, in turn, will lead to better neurological diagnosis and treatment.

Apart from clinical settings, this EEG-based diagnosis method has the possibility of making remote diagnostics possible. It has revolutionary potential, especially in emergency and distant care. In poverty-stricken areas, medical professionals will be able to instantly spot vital brain activity like seizures. By developing portable EEG devices with this method can speed up identification of seizures in ERs or ambulances, which will ultimately lead to faster interventions. Military and field applications can make quick brain assessments, even in remote locations (Fig. 1).

4 Methodology

4.1 Data Gathering and Preprocessing

This research used a broad dataset of EEG recordings covering a range of neurological disorders, such as seizures and non-seizures. It made sure that the dataset had correct labels indicating the presence or absence of seizures and other critical brain activities. In order to eliminate noise, artifacts, and baseline drift, preprocessed the EEG data by using methods like filtering, baseline correction, and artifact removal algorithms. Divide the EEG signals into suitable intervals, usually a few seconds, in order to make feature extraction and categorization easier [7].

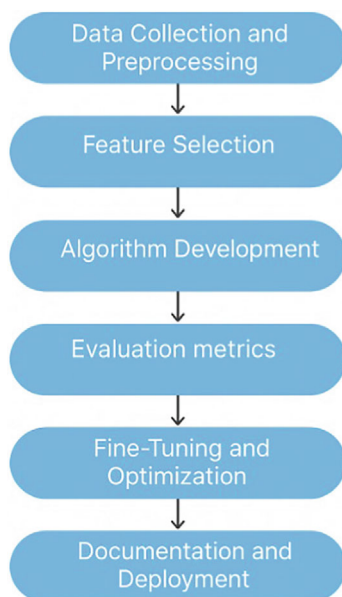


Fig. 1 Flowchart of implementation

4.2 Feature Extraction

For feature extraction, the model used EEG spectrogram images along with metadata and selected relevant features. The metadata consists of unique IDs for both the EEG recordings and spectrograms and time offsets to sync them. It also labels the different brain activities like seizures, which helps to focus on the important data for training (Fig. 2).

4.3 Algorithm Development

Created and applied machine learning models to automatically identify and classify vital physiological functions in EEG waveforms. In this research, three models were applied Visual Geometry Group 16 layers (VGG16), Residual Network (ResNet), and EfficientNet. These are convolutional neural network (CNN) models commonly used for image classification and object detection.

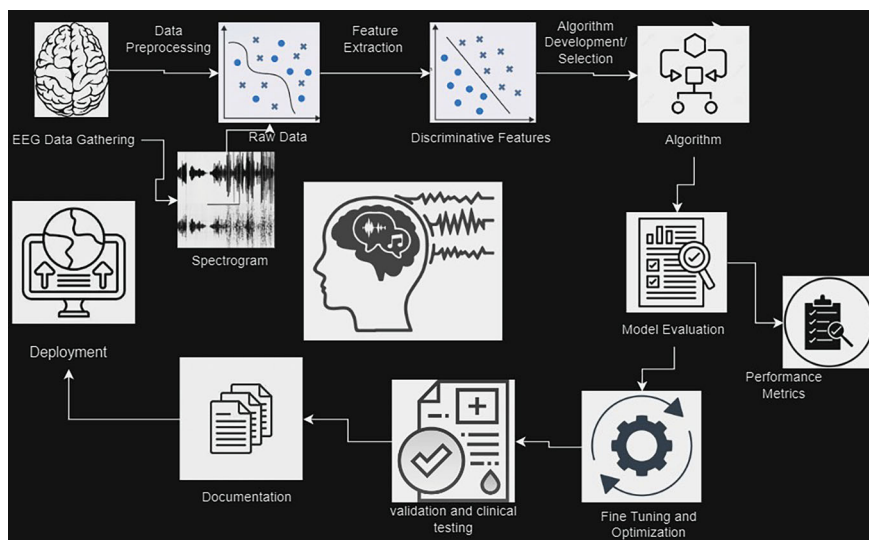


Fig. 2 Methodology

4.4 Model Evaluation

Analyzed the performance of the generated algorithms by utilizing relevant metrics, including precision, $F1$ score, and accuracy. Verify the algorithms' generalization potential and practicality by running them via independent test dataset validation. To illustrate the automated algorithms' effectiveness and possible therapeutic influence, compare their performance to that of current manual analysis techniques and human expert annotations.

4.5 Clinical Validation and Deployment

Work with clinical partners to confirm that the automated EEG analysis algorithms work as intended in actual healthcare environments. To evaluate the clinical value, effectiveness, and effect on patient outcomes of incorporating automated EEG analysis into standard clinical practice, prospective studies should be conducted. To evaluate the clinical value, effectiveness, and effect on patient outcomes of incorporating automated EEG analysis into standard clinical practice, prospective studies should be conducted. To enable the smooth integration of the automated algorithms into current EEG monitoring equipment and clinical procedures, create user-friendly interfaces and integration frameworks. Assure adherence to ethical and legal guidelines controlling the use of medical software tools, such as those pertaining to patient safety, data protection, and clinical validation requirements.

The aim was to create a reliable and executable technique for efficiently identifying and classifying crucial medical activities, particularly seizures and other brain patterns that lead to seizures, in EEG data.

5 Result

The devised method showcased promising results across various evaluation criteria for automated classification of important medical activities in EEG signals.

The performance of the developed model to detect critical brain activities was evaluated by analyzing the loss and accuracy curves generated during the training and validations stages. These curves show how well the model converges and generalizes to the dataset. Here, this research displays the learning curves of the best working model among the other models, i.e., EfficientNet.

5.1 *Loss Curve Analysis*

The loss curve showcases how the algorithm optimizes by reducing the loss function such as cross-entropy over each epoch. The loss curve consistently dropped throughout the training process. This indicates that the developed algorithm is able to minimize the gap between the predicted labels and actual labels. This steady decrease in the loss function suggests that the training procedure was effective in optimizing the model and leading it to an optimal outcome (Fig. 3).

5.2 *Accuracy Curve Analysis*

The accuracy curve depicts the percentage of accurately classified samples over epochs, and this indicates how well the algorithm performs in classification tasks. In this case, the accuracy curve gradually increases, indicating that the model is able to differentiate between seizure and non-seizure EEG patterns. The consistent rise in accuracy highlights the algorithm's ability to extract important features for accurate classification and gradually learn over successive epochs of training (Fig. 4).

5.3 *Interpretation and Implications*

The trends observed in both loss and accuracy curves suggest that the training and validation process of the algorithm were successful in building a high-performing

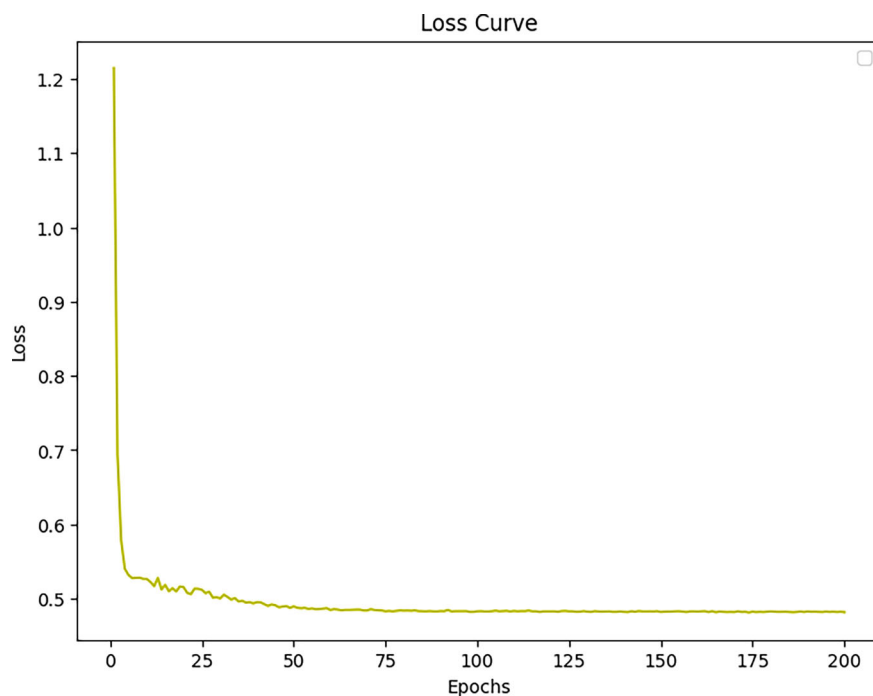


Fig. 3 Loss curve of EfficientNet. *Source* Kaggle Notebook

model for the classification of EEG signals. The model is able to learn informative characteristics from the spectrograms and make accurate predictions about harmful brain activities, particularly seizures (Fig. 5).

Moreover, the steadiness between the training and the validation curves (Fig. 6) indicates considerably less over-fitting. This shows that the model generalizes well to the unseen data. The impressive performance of the model is very important especially for clinical practices, as it enhances the model's efficiency to make accurate predictions. This will facilitate timely intervention and improve healthcare.

The analysis of loss and accuracy curves provides compelling evidence of the effectiveness and reliability of the developed automated EEG analysis algorithm. By leveraging machine learning techniques and signal processing algorithms, the algorithm demonstrates promising performance in detecting and classifying critical medical activities within EEG signals, paving the way for enhanced neurological diagnosis and treatment strategies.

Furthermore, the automated EEG analysis technique showed promise in real-world clinical settings for streamlining diagnostic processes and enabling prompt interventions. Healthcare practitioners can make better decisions and offer greater care if crucial medical activity, such as seizures, are identified sooner. People with

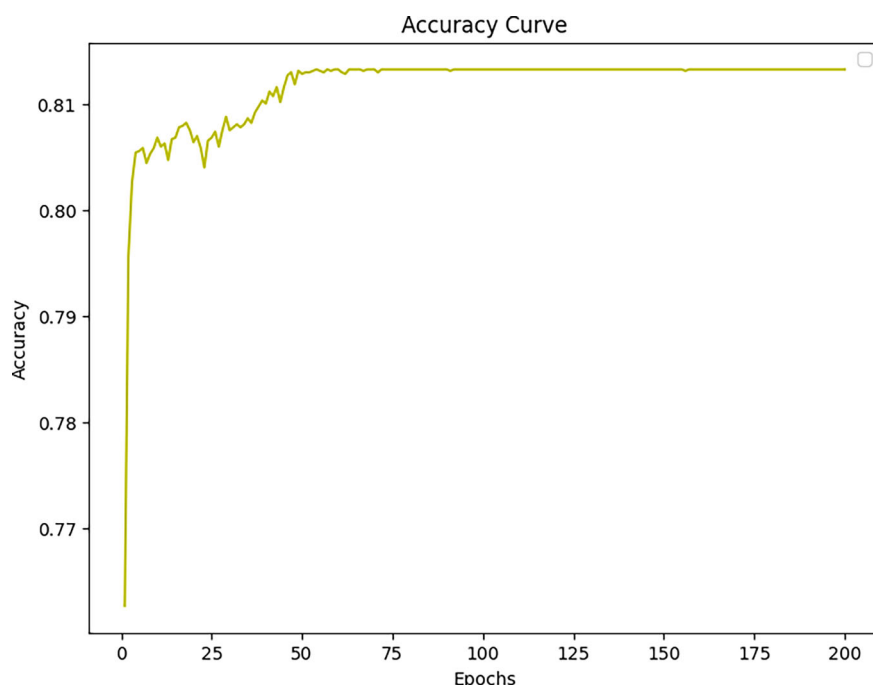


Fig. 4 Accuracy curve of EfficientNet. *Source* Kaggle Notebook

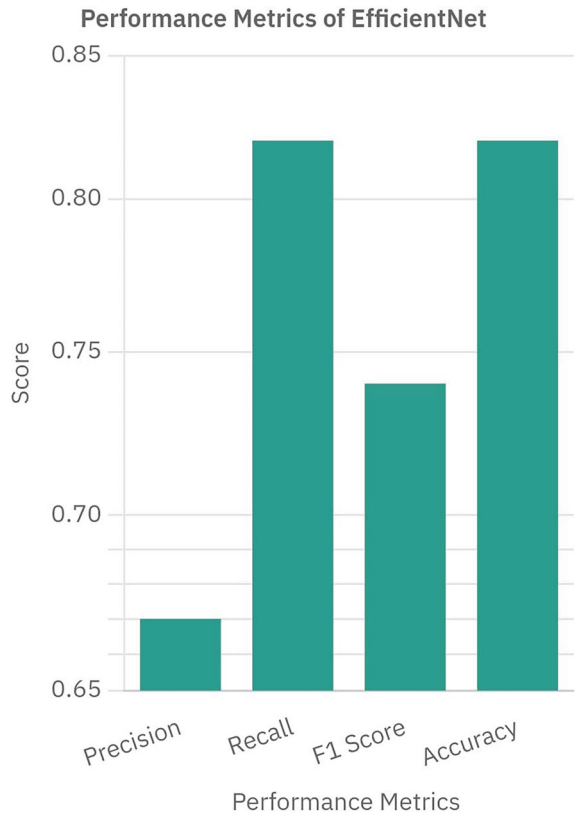
neurological problems will receive timely treatment. All things considered, the outcomes of the algorithm's application and assessment highlight how important it is to transform EEG analysis and advance the identification and treatment of neurological illnesses.

The EfficientNet algorithm regularly produced results that distinguished between seizure and other harmful brain activity states in spectrograms with better accuracy, *F1* score, recall and training and validation curves as compared to the other models. Thorough training on test datasets validated this accuracy, indicating the model's performance for generalization.

6 Discussion

The automated method for detecting and classifying essential medical activities in EEG signals showed promising results. The societal implications of this work are significant. By enabling fast and accurate detection of harmful brain activities, this algorithm supports early intervention, potentially reducing the burden on health-care systems. Moreover, this tool can extend EEG-based diagnostics to remote and emergency settings, where access to specialized neurodiagnostic resources may be

Fig. 5 Visualization of performance metrics of EfficientNet



limited. This capability underscores the broader utility of our algorithm as a scalable solution to neurological assessments in diverse healthcare environments.

In this research, three models were applied, Visual Geometry Group 16 layers (VGG16), Residual Network (ResNet), and EfficientNet. These are convolutional neural network (CNN) models commonly used for image classification and object detection. It was observed that ResNet took a longer time for the training process as compared to other models. Even with its ability to learn complicated features and deep architecture, some issues were encountered in enhancing the performance of the model.

Conversely, EfficientNet provided better results despite having a comparatively simpler architecture than ResNet. It has a balanced scaling approach that provides effective learning without compromising training time and efficiency. This highlights EfficientNet’s suitability for scenarios where computational resources such as processing units (CPUs and GPUs), sufficient memory and high-speed internet connections are limited.

VGG16 being a classic architecture with proven effectiveness showed a comparable performance to EfficientNet. However, the training process took slightly longer than EfficientNet and with quite higher computational demands.

6.1 Performance Metrics Comparison

In this section, the comparison between the performance metrics of EfficientNet and other models based on key evaluation criteria is shown.

Recall: EfficientNet achieved the highest recall rate among the models tested, underscoring its effectiveness in accurately identifying relevant instances. High recall rate means that the model is able to minimize false negatives accurately which helps to rely on the model as it ensures that all the cases are accurately detected and not overlooked.

F1 Score: EfficientNet also yielded the highest *F1* score. It is also considered as a harmonic mean between precision and recall hence maintaining a strong balance between them. *F1* score shows the correctness of the model in accurately classifying positive instances while maintaining a low false positive rate. It is commonly used to output reliable results in cases where precision and sensitivity both are essential.

Accuracy: EfficientNet gave the highest accuracy than Resnet and VGG16 demonstrating its ability to accurately classify medical images. The high accuracy of EfficientNet depicts the model’s ability of making correct predictions of the diagnosis, and it also depicts the model’s robustness in handling diverse datasets.

In conclusion: The performance metrics suggests that EfficientNet is the most effective and accurate model amongst VGG16 and Resnet. EfficientNet excels in accurately identifying relevant instances and maintains a balance between sensitivity and precision and hence demonstrates high classification accuracy (Table 1).

Training Loss and Validation Loss Curve: Initially, the loss curve drops sharply and then decreases gradually. This indicates that the model is learning and improving its accuracy over each epoch. Toward the end of the training process, both the curves come close to each other, showing that the model generalizes well to the unlabeled data. The training and validation losses both converge to a low value, and this shows that the model is performing well without any significant over-fitting.

Table 1 Performance metrics of EfficientNet

EfficientNet	Performance metrics
Precision	0.6768533097608694
Recall	0.8227109515260324
F1 score	0.7426885861350491
Accuracy	0.8227109313011169

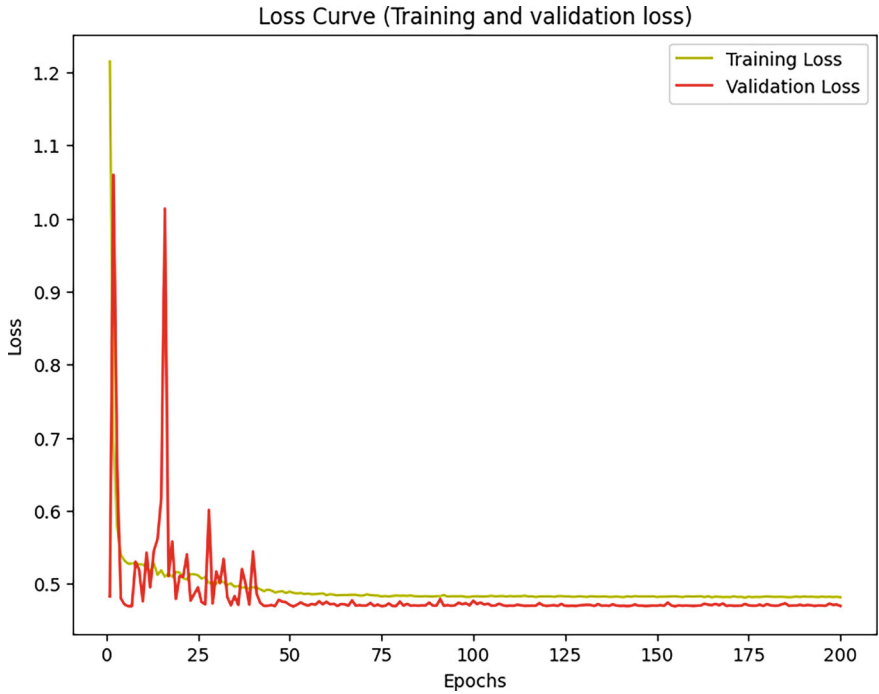


Fig. 6 Training and validation loss curve. *Source* Kaggle Notebook

Training Accuracy and Validation Accuracy Curve: Initially, there is a notable instability in the validation accuracy curve. After the initial volatility, the training accuracy appears to be stable, signifying that the model has accurately adapted to the unlabeled data well. The proximity of training and validation accuracy suggests that the model is generalizing well and is not over-trained. The accuracy values are relatively high (above 0.8), which indicates that the model is performing well on classifying the spectrograms (Fig. 7).

7 Future Scope

Future research will focus on fine-tuning the model’s architecture, optimizing hyper-parameters, and exploring various training techniques to further enhance performance. Additional goals include adapting the algorithm for real-time processing, enabling its integration in continuous EEG monitoring systems. Collaborating with healthcare stakeholders for broader validation across diverse EEG datasets will be

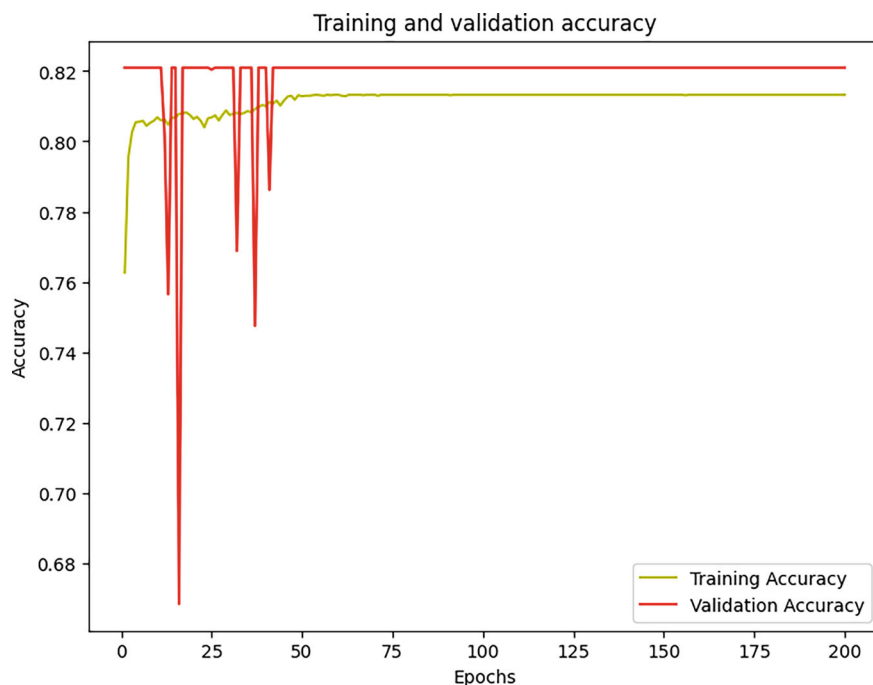


Fig. 7 Training and validation accuracy curve. *Source* Kaggle Notebook

essential for refining its application in clinical practice. Expanding the model's sensitivity to a wider array of brain activity patterns beyond seizures is also a potential direction for further research.

8 Conclusion

This study aims to assist doctors and brain researchers in identifying seizures and various types of brain activity that may lead to brain damage. With this approach, physicians can provide treatments more swiftly and accurately. The algorithm will support researchers in identifying seizures and harmful brain activities, potentially aiding the development of more effective treatments. The EfficientNet algorithm consistently demonstrated superior performance in distinguishing between seizure-related and other harmful brain activity states in spectrograms, demonstrating enhanced accuracy, $F1$ score, recall, and improved training and validation curves compared to

alternative models. Comprehensive training on both test and training datasets confirmed this accuracy, highlighting the model's performance in generalization. However, challenges such as dataset diversity, artifacts, and the variety of neurological conditions require ongoing research and development. Future research directions will focus on fine-tuning, optimizing, enhancing accuracy, enabling real-time monitoring capabilities, and fostering collaboration with healthcare stakeholders.

References

1. Nanthini BS, Santhi B (2017) Electroencephalogram signal classification for automated epileptic seizure detection using genetic algorithm. *J Nat Sci Biol Med* 8(2):159–166. https://doi.org/10.4103/jnsbm.JNSBM_285_16. PMID: 28781480; PMCID: PMC5523521
2. Cai L, Gao J, Zhao D (2020) A review of the application of deep learning in medical image classification and segmentation. *Ann Transl Med* 8(11):713. <https://doi.org/10.21037/atm.2020.02.44>
3. Suganyadevi S, Seethalakshmi V, Balasamy K (2022) A review on deep learning in medical image analysis. *Int J Multimed Inf Retr* 11(1):19–38. <https://doi.org/10.1007/s13735-021-00218-1>
4. Chan HP, Samala RK, Hadjiiski LM, Zhou C (2020) Deep learning in medical image analysis. *Adv Exp Med Biol* 1213:3–21. https://doi.org/10.1007/978-3-030-33128-3_1
5. A'izzah AVR, Rahmatulloh A, Purwayoga V, Improving harmful brain activity classification model with electroencephalography spectrogram using transfer learning. Available at SSRN: <https://ssrn.com/abstract=4928067> or <https://doi.org/10.2139/ssrn.4928067>
6. Habijan M, Šojo R, Tolić IH, Galić I (2024) Harmful brain activity classification using ensemble deep learning. In: *2024 International symposium ELMAR*, Zadar, Croatia, pp 109–112. <https://doi.org/10.1109/ELMAR62909.2024.10694603>
7. HMS Harmful Brain Activity Classification, <https://www.kaggle.com/competitions/hms-harmful-brain-activity-classification>
8. Nielsen M, Neural networks and deep learning. Publisher, Location
9. Harmful Brain Activity Classification of Spectrograms with Transfer Deep Learning - Scientific Figure on ResearchGate. Available from: https://www.researchgate.net/figure/AugmentedSmoothedSpectrograms_fig2_380025294. Accessed 12 May 2024
10. Siuly S, Li Y, Zhang Y (2016) EEG signal analysis and classification. *IEEE Trans Neural Syst Rehabil Eng* 11:141–144
11. Sharma M, Patel S, Acharya UR (2020) Automated detection of abnormal EEG signals using localized wavelet filter banks. *Pattern Recogn Lett* 133:188–194
12. Pathak D, Kashyap R, Rahamatkar S (2022) A study of deep learning approach for the classification of electroencephalogram (EEG) brain signals. In: Artificial intelligence and machine learning for EDGE computing. Academic Press, pp 133–144
13. Shah V, Golmohammadi M, Obeid I, Picone J (2021) Objective evaluation metrics for automatic classification of EEG events. In: Biomedical signal processing: innovation and applications, pp 223–255
14. Dereymaeker A, Matic V, Vervisch J, Cherian PJ, Ansari AH, De Wel O, Jansen K (2019) Automated EEG background analysis to identify neonates with hypoxic-ischemic encephalopathy treated with hypothermia at risk for adverse outcome: a pilot study. *Pediatrics Neonatology* 60(1):50–58
15. Ziyabari S, Shah V, Golmohammadi M, Obeid I, Picone J (2017) Objective evaluation metrics for automatic classification of EEG events. *arXiv preprint arXiv:1712.10107*
16. Al-Kaysi AM, Al-Ani A, Loo CK, Powell TY, Martin DM, Breakspear M, Boonstra TW (2017) Predicting tDCS treatment outcomes of patients with major depressive disorder using automated EEG classification. *J Affective Disorders* 208:597–603

17. O'Toole JM, Boylan GB (2019) Quantitative preterm EEG analysis: the need for caution in using modern data science techniques. *Front Pediatrics* 7:174
18. Pathak D, Kashyap R, Rahamatkar S (2022) A study of deep learning approach for the classification of electroencephalogram (EEG) brain signals. In: *Artificial intelligence and machine learning for EDGE computing*. Academic Press, pp 133–144
19. Thoduparambil PP, Dominic A, Varghese SM (2020) EEG-based deep learning model for the automatic detection of clinical depression. *Phys Eng Sci Med* 43:1349–1360
20. Zeng H, Yang C, Dai G, Qin F, Zhang J, Kong W (2018) EEG classification of driver mental states by deep learning. *Cognitive Neurodynamics* 12:597–606
21. Nakagome S, Craik A, Sujatha Ravindran A, He Y, Cruz-Garza JG, Contreras-Vidal JL (2022) Deep learning methods for EEG neural classification. In: *Handbook of neuroengineering*. Springer Singapore, pp 1–39